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# Real-Time Intelligence with Microsoft Fabric

## Empowering Data-Driven Decisions in the Era of AI

Johan Ludvig Brattås
& Frank Geisler

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## Real-Time Intelligence with Microsoft Fabric

by Johan Ludvig Brattås and Frank Geisler



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Production Editor: Katherine Tozer

Interior Designer: David Futato

Cover Designer: Susan Brown

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# Chapter 1. Introduction to Microsoft Fabric

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>
> With Early Release ebooks, you get books in their earliest form—the author's raw and unedited content as they write—so you can take advantage of these technologies long before the official release of these titles.
>
> This will be the 1st chapter of the final book. Please note that the GitHub repo will be made active later on.
>
> If you'd like to be actively involved in reviewing and commenting on this draft, please reach out to the editor at *rfernando@oreilly.com*.

Fabric is the new cool kid on the block from Microsoft. You might have read and heard a lot of hype about it, and for good reason. Fabric has been touted by Microsoft as the all-in-one solution for enterprises looking to streamline their data management and analytics needs. With a unified platform, enterprises can focus on deriving insights from their data instead of dealing with the unnecessary complexity of using disparate products and tools. In this chapter, we'll give you a high-level introduction to Fabric, including its core components, key features, data storage paradigms, and licensing options. All together, you'll see how Fabric is able to provide you with the tools for real-time intelligence. Let's jump right in and have a look at what Fabric is all about.

## Challenges of a multi-product data analytics platform

When you need data analytics and data management, there is a plethora of products on the market that can meet your needs. A custom solution utilizing products from different vendors, such as Airflow for orchestration and Jupyter Notebooks for transformation, will do the job but with a lot of moving parts and complex integrations.

If you build a data analytics platform from different vendors' products, you'll likely run into some challenges:

*Complexity Issues*

> Each vendor can implement things like security, licensing, or data handling differently. The more variety of moving parts there are, the more processes you have to manage.

*Integration challenges*

> Different vendors may use varying data formats, protocols, and APIs, which can lead to integration challenges. Ensuring that all components work seamlessly together often requires custom development and continuous maintenance.

*Licensing*

> Licensing options differ from company to company, like subscriptions or a one-off payment. It takes time and effort to keep track of these different vendor contracts.

*Management Overhead*

> Managing multiple products from different vendors can add complexity to configuration, monitoring, troubleshooting, and updating/maintaining the platform.

*Steep learning curve*

> Each vendor's product comes with its own set of tools, interfaces, and terminologies. This steepens the learning curve for users and

administrators, necessitating more training and knowledge management and a bigger team.

*Security risks*

Different vendors may have varying levels of security standards and protocols. Ensuring consistent and comprehensive security across all components can be challenging, and potentially can lead to vulnerabilities.

*Complex compliance management*

It becomes complicated to maintain compliance with industry regulations—like GDPR or HIPAA—while handling many vendors, all with their individual compliance practices and documentation requirements.

*Troubleshooting*

When issues arise, coordinating support across different vendors can be time-consuming, difficult and complicated. Each vendor may need to be involved in diagnosing and resolving issues, leading to potential delays and an enormous overhead.

*Blame game*

If issues arise, people may blame each other, which does not lead to progress and complicates and prolongs the troubleshooting process. This will lead to frustration and ineffectiveness in resolving the problem.

In our experience, sourcing all the platform components through a single vendor minimizes (or in some cases eliminates) these particular challenges, which is where Fabric comes into play.

# Fabric Key Features and Capabilities

Microsoft Fabric is an end-to-end data analytics platform that integrates various data services, including data engineering, data integration, data warehousing, and data science, into a unified environment. As a Software-as-a-Service (SaaS), it simplifies the infrastructure management, the orchestration and integration of the various compute engines etc. It aims to streamline data management and analytics processes, enabling organisations to derive insights more efficiently and effectively.

Fabric is made up of services called "experiences". Each experience contributes to building an analytical solution, starting with Data Ingestion and ending with Data Visualization and Reporting.

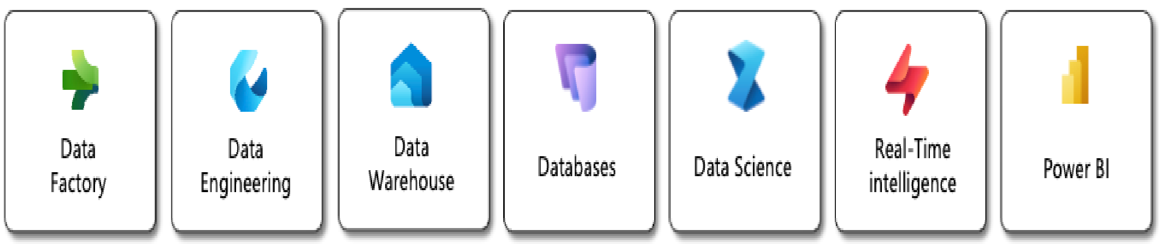



*Figure 1-1. The different experiences in Microsoft Fabric*

Fabric offers the following services, also shown in Figure 1-1:

*Data Factory*

One of the most important features needed in data analytics is data integration, because if you don't have data, you can't do analytics very well. The Data Factory experience provides various capabilities designed to connect to a variety of data sources and land the data in Fabric. It provides high compatibility with and out-of-the-box connectors for a variety of data sources, from on-premises databases to cloud storage services and SaaS applications to third-party platforms. It supports both batch, change data capture (CDC) and real-time data ingestion methods for a wide range of use cases. You can decide to run Extract-Transform-Load (ETL) or Extract-Load-Transform (ELT) processes based on your requirements and architecture. You can also use Power Query and the like to support data transformation, data wrangling and processing to help you clear, shape, or enrich information for analysis.

> **NOTE**
>
> Although the Data Factory experience has the same features as Azure Data Factory, they are not the same service.

*Data Engineering*

The Data Engineering experience enables you to design, develop, and manage the infrastructure required to collect, store, process, and analyze large volumes of data. Where the Data Factory experience focuses on traditional ETL/ELT, the Data Engineering experience focuses on a data lakehouse architecture approach with Spark and Notebooks. Notebooks in Fabric support Scala, PySpark, Spark SQL and R.

*Data Warehouse*

In Fabric there are two takes on the classic Data Warehouse approach: the Data Warehouse itself, which is a classic online analytical process (OLAP) massively parallel process (MPP) database, and another one called the Lakehouse. The Lakehouse is used for the Data Engineering experience, while the Data Warehouse is used for T-SQL and databases. We will discuss Data Warehouse and Lakehouse in detail later in this chapter.

> **NOTE**
>
> The Fabric Data Warehouse is not a replica of the Synapse dedicated SQL Pool service. Rather, it is a new service.

*Databases*

While the Data Warehouse is an OLAP MPP service, there are times when you might need a traditional transactional SQL database. The Database service allows you to create traditional SQL databases within

Fabric. You can use them for hosting metadata for your pipelines, as backend storage for your AI apps or a simplified approach to host other transactional data.

*Data Science*

The Data Science experience within Fabric is powered by capabilities from Azure Machine Learning Studio and other Azure services. It allows you to discover and manipulate large datasets, execute complex transformations, do experiments and build predictive models, based on the data you make available in OneLake. You can work in a full-code environment using notebooks, or take the low-code approach as you might be familiar with if you have done any work in Azure ML Studio.

*Real-Time Intelligence*

The Real-Time Intelligence experience, which is the main focus of this book, enables real-time insights and actions against live data streams as well as historizing events and time-series data for further analysis.

*Power BI*

With the Power BI experience, the data that has been processed through Microsoft Fabric can be transformed into meaningful insights that you can use to drive strategic decisions using Power BI.

Now, let’s take a closer look at how the services mentioned across all these experiences are enabled in the overall architecture of Fabric.

## High level architectural overview

Fabric is engineered to offer an integrated path for data storage, processing, analytics, and orchestration at any scale and across diverse environments. In this section, we’ll show you the high-level architecture of Fabric and how the parts fit together.

Fabric is made of several technical core components that provide all the functionality of Fabric that you can use to create your analytics solution. The front-end GUI is where users normally interact with Fabric. The backend is where the compute engines and common data store reside. The REST APIs allow you to use tools other than the browser, and they also allow you to automate processes. The common metadata platform stores and manages all of the metadata on the platform. Lastly, we have security and governance elements. These components are shown in Figure 1-2:

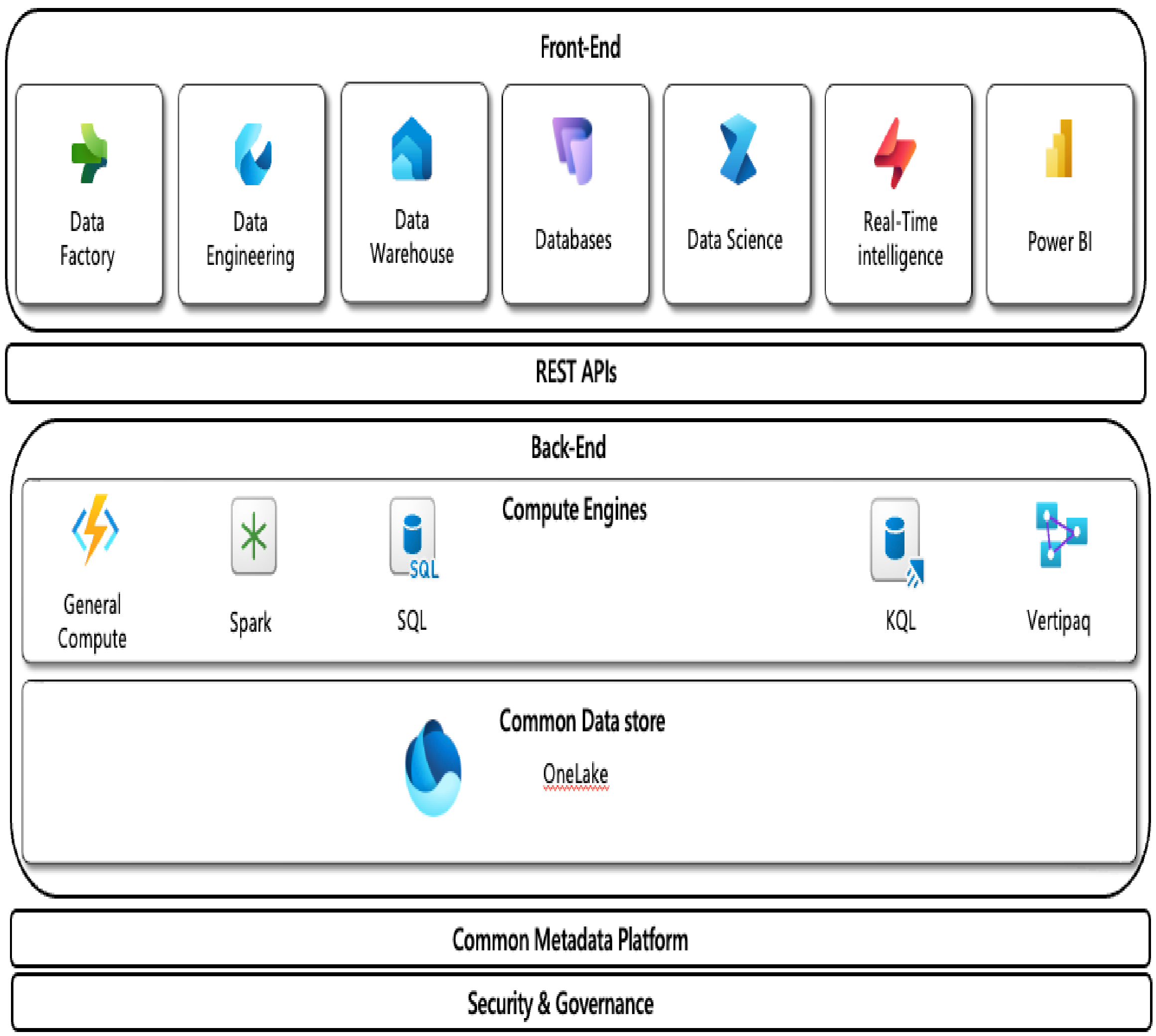



*Figure 1-2. High-level architecture of Microsoft Fabric*

We've already looked at the front-end experiences. Let's take a closer look at the other components:

- Common Data Store

- OneLake is the centralized data storage layer of Microsoft Fabric. It is designed to serve as the unified repository for all data types across an organization. It provides a scalable, cloud-based storage solution that integrates with Microsoft's data ecosystem.
- OneLake also supports advanced features like data versioning, so you can track changes and maintain historical data states.
- Compute Engines
- The serverless compute engines drive every processing, analysis, and transformation of data efficiently within the Microsoft Fabric. There are 4 main compute engines that you will have at your disposal when working in Fabric:

  *General Compute*

  > To handle all the compute that is not directly related to a data compute engine. For example, orchestrating pipelines, copying data, triggering Reflex actions or Data Functions.

  *SQL Engine*

  > There are three different SQL engines in Fabric: The Warehouse SQL engine, the SQL Analytical Endpoint and the SQL database engine. Basically, all are based on your familiar SQL Server for relational data processing. They provide you with high performance in querying, data manipulation, and complex joins, which makes it great for operational databases or data warehouses.

  *Spark Engine*

  > It is developed for processing huge data that makes it popular for big data transformation, aggregation, and machine learning tasks in a very professional way. It's particularly helpful when handling massive data sets and running distributed processing jobs.

*KQL Engine*

> This is a time series, low-latency processing engine that assists organizations in analyzing and acting upon data in real-time. We will be talking about the KQL engine throughout the book..

*Vertipaq*

> Power BI's embedded compute (also used in Azure Analysis Services) can run rapid in-memory analytical querying, data processing and visualization. It allows users to engage with data and create reports and dashboards that reflect up-to-the-minute insight.

These compute engines are naturally engineered to work seamlessly within the Microsoft Fabric ecosystem to provide users with the power of choice for the best tool for their particular data processing needs.

Whatever you want to do, the important part here is just being consistent and having a plan on how to build your version of Fabric. Sprinkling one solution here and another solution there makes the whole system very hard to maintain.

- Metadata Platform
- To be able to expose all the data in OneLake across the various compute engines, Fabric needs to have a shared metadata store. This is shared on the tenant level. We'll talk about tenants later in this chapter.
- REST APIs
- All communication between the back-end, metastore and front-end components is handled by a set of REST APIs.
- Security and Governance

- Security and data governance are important building blocks in any data platform. And the main focus areas are protection, integrity, and compliance of data through its entire lifecycle. As mentioned previously, Fabric is built on top of the existing Azure security features such as encryption, Identity Access Management (IAM) with multi-factor authentication based on Entra ID, security on workspace or data level down to cells in tables via SQL Analytics endpoints.
- Microsoft Purview is integrated with Fabric and allows you to scan, categorize and control data on your platform. Purview also offers additional services such as Information Protection, Data Lifecycle Management, Data Quality and Data Loss Prevention.

As you can see, Fabric is a comprehensive, modern data platform that allows you to analyze data in a variety of ways. Now, how you store your data for analysis impacts how you use it in Fabric. Let's look at the data storage paradigms available to you in Fabric.

## Data Storage Paradigms

When it comes to data analytics, how you store, organize, and access data impacts your organisation's ability to process and analyse data efficiently, affecting performance, scalability, integrity, and security. You must consider how you manage data in terms of volume, variety, and speed. Different data storage paradigms offer benefits for specific needs, such as the amount of data written to storage or read from storage, the speed of those transactions or how it handles a variety of file types and file sizes. Fabric is lake-centric. This means that data is stored in storage decoupled from compute, as opposed to, say, a relational database. Additionally, Fabric relies on what is called a lakehouse format - Delta Lake - which is an attempt to impose transactional database functionality, such as ACID (Atomic, Consistent, Isolation, Durability) principles, on data lake file storage. In Delta Lake terms, we talk about a transactional log and metadata in combination with the parquet files that store the data. This is also why you will see in

Microsoft documentation the phrase Delta Parquet used - even though the format is called Delta Lake.

That said, Fabric offers three data storage paradigms: data warehouses, data lakehouses, and eventhouses. Understanding these paradigms helps you choose the best data architecture to meet your goals and operational needs.

## Data Warehouses

A *data warehouse* is a centralized repository that stores large volumes of structured data from various sources, designed for query and analysis. It enables your organisation to consolidate data, perform complex queries, and generate reports, facilitating informed decision-making. Data warehouses are optimised for fast retrieval and analysis, supporting business intelligence activities by providing a consistent and reliable view of historical and current data.

However, the work needed to make sure data is of the right quality and is correct and on top of that making sure data from a variety of source systems can be joined together and queried at the same time can be time consuming. Unfortunately, the specialized team working on building and running a data warehouse can be seen as a bottleneck.

The key features of a data warehouse are:

*Centralised Repository*

Stores large volumes of structured data from various sources.

*Optimised for Query and Analysis*

Designed for complex queries and reporting.

*Historical Data Storage*

Maintains historical data for trend analysis and decision-making.

*Consistent Data*

Provides a reliable and consistent view of data.

*Data Integration*

> Combines data from multiple sources into a unified format for cross-system querying.

With the advent of “big data” (increase in velocity, volume and variety of data) the change of volume, variety and velocity of data strained the relational databases that the traditional data warehouse were running on. On top of that, the specialized team also struggled with the loads. Enter the data lake - our next storage paradigm - which tries to solve the problems of the data warehouse.

## Data Lakehouses

*Data lakes* allows for storing large volumes of diverse data types without predefined schemas, enabling flexible data exploration and advanced analytics. Without the need to work on data quality, normalizing data and enabling cross-system querying - data is made available in the data lake much faster than in a data warehouse.

However, this transfers the cost of transforming, qualifying and aligning the data over to the users. And on top of that, since each user now is able to reinvent the wheel so to speak - your organization suddenly might find itself with multiple versions of the truth.

These issues have led to data lakes being a paradigm most organizations move away from - and it has been a trend since 2019

The following are the key features of a data lake:

*Raw Data Storage*

> Stores data in its native format, including structured, semi-structured, and unstructured data.

*Scalability*

> Can handle large volumes of diverse data types.

*Flexibility*

Allows for flexible data exploration and analysis without predefined schemas.

*Real-Time Processing*

Supports real-time data processing and analytics.

*Cost-Effective*

Provides a scalable and economical solution for managing big data.

*Advanced Analytics*

Enables machine learning and advanced analytics applications.

*Data Variety*

Accommodates a wide range of data sources and formats.

So the shift in 2019 was when the third of our data storage paradigms was launched: The data lakehouse.

As the name says, it merges the best of traditional data warehousing and data lakes together in one consistent platform. The hybrid approach solves a lot of the limitations intrinsic within traditional systems for managing data and gives an organization much greater flexibility, scalability, and cost-effectiveness in storing and analyzing vast diverse data volumes.

A data warehouse is fine-tuned only for structured, schema-on-write data processing and querying while data lakes are designed to hold large amounts of semi- and unstructured schema-on-read data. A data lake traditionally is designed to let business build point solutions on top of the data - often leading to silos as well as challenges with data quality and correctness since governance typically has been lacking.

The main strength of a data lake is enabling self-service analytics and data science projects, The data lakehouse offers a strong, multi-role solution for organizations seeking a unified data architecture. It combines the advantages of a data lake and a data warehouse to provide better management, support for a wide array of analytics, and additional flexibility and scalability. As more and more complex data gets generated and consumed by business, data lakehouses will help in eliciting actionable insights from that data and staying ahead of the market competition.

Here are the key features of a data lakehouse:

*Unified Storage*

Combines the capabilities of data lakes and data warehouses in a single platform.

*Structured and Unstructured Data*

Supports storage and processing of both structured and unstructured data.

*Scalability*

Offers scalable storage and computing resources.

*Real-Time Analytics*

Facilitates real-time data processing and analytics.

*Data Management*

Provides tools for data governance, quality, and lifecycle management.

*High Performance*

Optimised for fast query performance and data retrieval.

*Advanced Analytics*

Supports machine learning, AI, and complex analytical workloads.

*Cost Efficiency*

Balances cost-effective storage with decoupled high-performance computing.

## Eventhouses

In Microsoft Fabric, there is also the Eventhouse, which is the front platform to capture and manage different sources of event data. An Eventhouse is a different proposition from any conventional data storage solution, which is focused on static data. It is designed for highly dynamic and time-sensitive streams - or time series data. It provisions the infrastructure required for handling large volumes of event data upon arrival, hence enabling real-time analytics, monitoring, and decision-making.

The key features of an Eventhouse are:

- Real-time data ingestion
- Stream processing
- Scalable, high-volume architecture
- Event-driven automation
- Historical time series storage

Eventhouses will be described more in detail later in this book

## Choosing the right solution for your needs

Because Fabric offers three storage paradigms, and at the same time uses the shared metadata and common data store to allow sharing data across paradigms - you can end up in a situation where you use all of them. And that might be OK for your organization, however - we do recommend that you consider specifically what kind of data your organization requires, what use cases it needs to serve, and what the strategic goals are. Each of these architectures does a particular, highly specialized task in an optimal way for different kinds of data processing and analysis. Knowing their strengths and applications will help in choosing the right solution for your needs.

Your organizational data characteristics, workload requirements, and business objectives would base this choice of architecture on:

- Choose a Data Warehouse in cases where structured data and reliable long-term reporting are required. Additionally if your organization is focused on SQL, this is a natural choice.
- Consider a Data Lakehouse when looking for a supple, unified platform that handles a broad array of data types and supports historical and advanced analytics. This will also include working in other languages than pure SQL, such as R or Python.
- Pick an Eventhouse when you need instant data processing and action on it immediately; this happens mostly in scenarios that require real-time responsiveness. Most eventhouse services use SQL, but you might also see KQL used.

The right choice can sometimes involve more than one of these, to fit your data strategy and business needs. Knowing the powers and applications of Data Warehouses, Lakehouses, and Eventhouses enables one to make a very informed decision that will align with goals set by an organization. As you might have noticed, we did not mention a pure data lake among the selections above. Simply because Fabric would be overkill in a solution that just demands a scalable data lake storage and a way to land data into the data lake.

## Licensing and capacities

As SaaS Analytics platform, Fabric has what appears as a very simple license model. You purchase a given capacity - which is what Microsoft calls their black box of compute+ram++ and off you go. However, since Power BI and Fabric co-exist in the same space, you need to think about both what Power BI licenses you need as well as what capacity size you need to be able to run the various services of Fabric and Power BI.

This section will give an overview of Fabric's licensing and capacity model: the tenant, capacity, and workspace setup. We'll also discuss what to consider when choosing a license based on your organization's size, industry, and specific use cases.

## Tenant

Microsoft Fabric is built on top of a Microsoft EntraID tenant. That tenant is bound to one Domain Name System (DNS). So if your organization already has Microsoft services such as Microsoft365 or Azure you already have a tenant established. But suppose you do not already have a Microsoft Entra tenant. In that case, you can either associate your domain with an existing tenant, or when you sign up for a Microsoft online service, a free, trial, or paid license a new tenant will be created automatically - the default tenant name is *<your organization name>.onmicrosoft.com*. After your tenant has been created you can start to add services such as Fabric or Power BI. And since Fabric capacities are Azure services you will need both a Power BI/Fabric portal as well as an Azure portal where the Fabric capacity resides.

## Workspace

Workspaces are containers within Fabric tenants, designed to house various Microsoft Fabric artifacts. Every user has access to a personal workspace called My Workspace. Additionally, common workspaces can be created to facilitate collaboration among team members or even company-wide collaboration. In Fabric, you will normally use different workspaces to function as environments such as Development, Test, and Production. One capacity can run on multiple workspaces, or be assigned to just one workspace. But it must be assigned to a minimum of one workspace - unless it is a Fabric CoPilot capacity, but that's an exception.

In any Fabric tenant, regardless of how many Fabric capacities you have, you will still have access to a shared capacity that hosts all the independent My Workspace and workspaces using the Pro or Premium Per User (PPU) licensing modes.

Workspaces can be created in, or assigned to, Microsoft Fabric capacities and you will be given an option to choose license mode when creating a new workspace. The workspace license mode determines what kind of capacity the workspace can be hosted within. How much a user can do within a workspace is also determined by its Workspace license mode.

*Table 1-1. Overview of Workspace licensing modes*

| Workspace license mode | User capabilities | Access | Supported experiences |
|---|---|---|---|
| Pro | Use basic Power BI features and collaborate on reports, dashboards, and scorecards. | To access a workspace with a Pro license mode, you need a Power BI Pro, Premium Per-user (PPU) license, or a Power BI individual trial. | Power BI |
| Premium per-user (PPU) | Collaborate using most of the Power BI Premium features, including dataflows, and datamarts. | To access a Premium Per User (PPU) workspace you need a PPU license or a Power BI individual trial. | Power BI |

| Workspace license mode | User capabilities | Access | Supported experiences |
|---|---|---|---|
| Premium per capacity (P SKUs) | Create Power BI content. Share, collaborate on, and distribute Power BI content. Deprecated - can no longer be purchased. | To create workspaces and share content you need a Pro or PPU license. To view content, you need a Microsoft Fabric (Free) license with a viewer role on the workspace. If you have any other role on the workspace, you need a Pro or a PPU license, or a Power BI individual trial. | All Fabric experiences |
| Embedded (A SKUs) | Embed content in an Azure capacity. | To create workspaces and share content you need a Pro, Premium Per User (PPU) or a Power BI individual trial license. | Power BI |

| Workspace license mode | User capabilities | Access | Supported experiences |
|---|---|---|---|
| Fabric capacity (F SKUs) | Create, share, collaborate on, and distribute Fabric content. | To view Power BI content with a Microsoft Fabric free per user license, your capacity must reside on an F64 or larger SKU, and you need to have a viewer role on the workspace. | All Fabric experiences |
| Trial | Try Fabric features and experiences for 60 days. | Microsoft Fabric (Free) license | All Fabric experiences |

## Capacity

In Microsoft Fabric, a capacity is a distinct pool of resources - a collection of compute and memory (as well as some limitations on storage) that is available to you at any given time. The resources available within a capacity are determined by its size, which ranges from 2 to 2048 capacity units. This dictates the level of computational power available for Microsoft Fabric. When purchasing a capacity you are asked to pick a SKU level - i.e. F2 or F64. The number after F is the capacity units you get,

A capacity is needed for the required infrastructure to run Microsoft Fabric. Your capacity gives you the option to create all the different services and experiences you want in Fabric. However, if you choose an F2 there are naturally more limitations on what you can do in Fabric compared to if you choose an F128.

You can have as many capacities as you want, and the price you pay for an F64 is the same as if you have 2 F32 or 4 F16.

When planning your Fabric environment it is important to select the right SKU for your organization's needs and use cases. Estimating what SKU level you need is not easy, though Microsoft has made a capacity calculator available so you can get a rough estimate at least.

You can find the official price list on Microsoft. Make sure you check out which regions are available to you, and the price for each region, as there are price differences.

As you can see, licensing and capacities in Fabric are not entirely as simple and straightforward as marketing might want you to believe. And it is a topic that we could spend a whole chapter on in another book. But we will cover the important capacity consumption mechanisms of the various components of Real-Time Intelligence through the various chapters of this book.

## Conclusion

We can't do Microsoft Fabric justice in just one chapter, but we hope that it has warmed you up for the rest of the book. In this chapter, we went through the core concepts and key components of this integrated data analytics platform as-a-service. You learned about the various experiences in Fabric, the core architecture as well as the data storage paradigms that Fabric is made to support. You also learned how licensing and capacity works.

As you saw in this chapter, Fabric's unified platform provides everything you need for data analytics, including Real-Time Intelligence. But, there's also an experience for business intelligence. What's the difference between the two? The next chapter will delve into what is meant by real-time intelligence and how it differs from business intelligence.

# Chapter 2. Business Intelligence Versus Real-Time Intelligence

> **A NOTE FOR EARLY RELEASE READERS**
>
> With Early Release ebooks, you get books in their earliest form—the author's raw and unedited content as they write—so you can take advantage of these technologies long before the official release of these titles.
>
> This will be the 2nd chapter of the final book. Please note that the GitHub repo will be made active later on.
>
> If you'd like to be actively involved in reviewing and commenting on this draft, please reach out to the editor at *rfernando@oreilly.com*.

Before we get into how to use Fabric for real-time intelligence (RTI), it's useful to understand how RTI differs from business intelligence (BI). While BI and RTI have many similarities, they serve distinct purposes and offer different advantages. This chapter aims to clarify the key differences between BI and RTI, helping you understand their unique roles in business strategy and operations.

## Business Intelligence

Business Intelligence (BI) refers to the methodologies, technologies, applications, and practices for collecting, integrating, aligning, and analyzing business information. BI as we know it came about from the introduction of data warehousing in the late 1980s. Since then, BI revolutionized the way businesses were run. Instead of acting on gut-driven decisions, BI helped people to make decisions on hard data. As more and more businesses became digital, the data made available via data warehousing and reporting enables leaders to react to tangible insights and trends.

Today, advanced analytics, micro-batch dataflows and user-friendly visualization are incorporated into modern BI solutions, such as PowerBI and Tableau, to provide technical and non-technical users at all levels with AI-driven insights.

The Key Features of BI are:

*Historical Data Analysis*

BI focuses on analyzing historical data to identify trends, patterns, and insights that inform strategic decisions.

*Cross-system querying*

BI - or the supporting data warehouse - collects and aligns the diverse systems in an organization in a way that allows decision makers to see how the business is doing without interacting with and understanding several systems.

*Batch Processing*

Batch processing refers to the method of transferring data from a source to a destination in discrete chunks or sets, rather than continuously or in real-time. Batch loading is typically scheduled to occur at regular intervals, such as hourly, daily, or weekly, and at these intervals selects the data that has been added or changed in the given interval and then loads it all at once into the data warehouse. It is worth mentioning that some systems are designed so that it is impossible for the batch process to know what data to select, and we have to fetch all the data every time and merge in the data warehouse.

*Dashboards and Reports*

BI tools provide dashboards and reports that summarize data, making it easier for stakeholders to understand and act upon the insights.

BI is best for gaining historical insights, such as identifying sales trends and customer behavior, monitoring key performance indicators (KPIs), and conducting market research and competitive analysis. It is also useful for enhancing operational efficiency by tracking production rates or minimizing equipment downtime. Likewise it can help improve resource allocation by analyzing cost data to identify areas where resources are being underutilised or overutilised to give some examples.

## Real-Time Intelligence (RTI)

Real-time intelligence refers to the capacity to process data, analyze it, and act on that information as it is generated - or as close to the generation as possible, thereby giving organizations the capability to gain immediate insight and to enable time-sensitive decision-making. Unlike traditional business intelligence, which rely heavily on historical data, real-time intelligence focuses on live data streams capturing events, detecting patterns, and uncovering opportunities or risks as they arise. The ability to do this is supported by a host of technologies, including event stream processing, time series databases, real-time dashboards, and machine learning to create a tightly integrated view.

Key features of RTI are:

*Instantaneous Data Processing:*

RTI systems process data in real-time, allowing businesses to react to events as they happen.

*Continuous Data Flow:*

RTI systems handle a continuous stream of data.

*Immediate Insights:*

RTI tools provide immediate insights and alerts, facilitating quick decision-making and responses.

*Event-Driven Architecture:*

RTI often uses an event-driven architecture, where specific events trigger data processing and analysis.

RTI is best for gaining immediate insights and enabling time-sensitive decision-making. It allows companies to react quickly to events but also to predict trends and optimize operations.

# Key Differences Between BI and RTI

Although both BI and RTI use data to support better decision-making, they differ in purpose and process. They differ in methods of processing data, the timeliness of data access, the kind of decisions they support, and the reactive versus proactive support they offer.

## Data Processing

BI typically operates on batch systems, which naturally induces some delay between when data is collected and when it is analyzed. A batch-mode operation is sufficient for cases that require historical perspectives to realize and make decisions. On the contrary, real-time intelligence uses data in an online mode, performing analysis on information generated by data sources almost instantaneously. Consequently, real-time processing allows for immediate insights, which can be essential in cases that require an immediate response, like fraud detection or system monitoring.

## Data Freshness

BI and RTI differ regarding the freshness and currency of the datasets they query. BI is carried out on historical data, giving a retrospective view that may never correspond to the realities of today. Even though modern BI tools are usually quite effective in forecasting long-term trends and patterns, they lack the immediacy upon which many rapid-response decisions depend. RTI, however, deals with live data streams and provides an instant and unchanged representation of ongoing events. This ability to express the current status of events gives RTI great value in situations characterized by extreme dynamism.

## Decision-Making Focus

BI is concerned with long-range, strategic decisions. It does an excellent job of identifying historical trends, viewing performance over time, and contributing to high-level business strategies. Meanwhile, RTI is meant for immediate tactical decision-making. By bringing information up to the minute, RTI allows firms to act rapidly regarding opportunities and challenges, making it critical for operational decision-making in matters such as inventory management or IT incident resolution.

## Complexity and Cost

BI and RTI have entirely different specifications regarding technical requirements. BI is typically less complicated and less expensive as it does not need the advanced infrastructure required for real-time processing. Thus, it's a strategy that lends itself to organizations wanting to analyze data without significant upfront technology investments. However, RTI needs a much more sophisticated setup able to handle high-velocity data streams and provide low-latency processing. Hence, RTI is at once more expensive and complicated than BI but provides huge value to organizations with a need for real-time insights to remain competitive.

In short, although BI and RTI share a lofty goal - that of using data to support decision-making - they adopt very different methodologies and applications, thus being suited for different situations. You must evaluate your organization's specific needs—whether to perform historical analysis with a view for strategic planning or to extract real-time insights that would allow rapid action—to select the most appropriate approach. That said, you don't need to use only one. BI and RTI support different needs, after all.

## Integration of BI and RTI

The best way for your organization to become data driven and get a competitive edge is to integrate both BI and RTI into your decision-making process. The two approaches strike a balance between strategic planning and operational agility, addressing both the "big picture" and the immediate actionable scenarios having the complementary strengths of BI and RTI.

The combination of information to support decision-making and, thus, allow for fine-tuning or modification of strategies based on insights from real-time data is indeed an unstoppable force.

## Use Cases and Industry Applications

So now that you have learned more about what RTI and BI is and isn't, let's take a look at how they are used in real-life applications, BI has been around for over 30 years and is already used across industries, so we won't spend much time discussing it here and will instead focus on RTI.

### Business Intelligence Use Cases

The following are some examples of use cases where BI makes a difference:

*Sales Performance Analysis*

> BI tools can analyse sales data to identify trends, track performance against targets, and uncover opportunities for growth. By visualising sales metrics, businesses can make informed decisions to boost revenue. For instance, you can recognise sales patterns over time, monitor sales against targets, understand different customer groups and evaluate product performance.

*Financial Reporting and Analysis*

BI enables detailed financial reporting and analysis, helping organisations manage budgets, track expenses, and optimise financial performance. It provides insights into profitability and cost management. This way you can track and manage budgets effectively, monitor and control expenses and identify cost-saving opportunities to mention a few examples.

*Customer Insights and Behaviour Analysis*

BI tools analyse customer data to understand behaviour, preferences, and satisfaction levels. This helps businesses tailor their marketing strategies and improve customer experience to understand customer actions and preferences, measure customer satisfaction, tailor marketing campaigns, develop strategies to retain customers and more.

*Operational Efficiency Improvement*

Identifying inefficiencies in business processes and operations is another BI area, enabling organisations to streamline workflows and improve productivity. It helps in resource allocation and process optimisation. This allows you to evaluate and improve business processes, optimise the use of resources, monitor employee productivity and streamline workflows.

*Market and Competitive Analysis*

Analyzing market trends and competitor data with BI tools help your business stay competitive. It provides insights into market dynamics and helps in strategic planning. You can easily track market trends, do competitor benchmarking, get a better understanding of market changes and spot market opportunities.

Business Intelligence can be applied in various use cases such as sales performance analysis, financial reporting, customer insights, operational efficiency improvement, and market analysis. Each use case leverages BI tools to provide actionable insights, helping organisations make informed decisions and optimise their performance.

## Real-Time Intelligence Use Cases

The following are a few RTI use cases to give you some hints of how your organization can make use of real-time intelligence:

*Real-time Monitoring and Alerts*

Monitoring constitutes one of the most known applications of real-time analytics. Systems set up with defined limits or characteristics of anomalies automatically triggers alerts that allow instant reaction from operations teams for any problem. For instance:

- IT Operations: Real-time monitoring of server performance, network traffic, or application logs ensures smooth maintenance of systems. Server downtimes, traffic bottlenecks, or security breach alerts can all be spotted and be handled when using real-time systems.

- Environmental Monitoring: You can keep track of environmental conditions such as temperature, humidity, or air quality, and respond in a timely fashion whenever major changes or problems occur.

*Predictive Maintenance*

With real-time intelligence, you can predict and preempt equipment failure by examining the sensor data continuously. This leads to a shift from scheduled maintenance to a predictive approach - which is more cost-effective. For example:

- Manufacturing: By predicting maintenance within factories, early signs of wear and tear in machinery ensure its operation as smoothly as possible.

- Transport: Airlines and logistics companies can exercise real-time data from vehicles, trains or aircraft in order to schedule maintenance before breakdown occurs.

*Fraud Detection and Security*

Real-time analytics is an essential tool for spotting suspicious activities or factors regarding fraud, therefore enabling fast responses from organizations. For example:

- Banking and Finance: Checking transactional data for patterns of fraud - e.g. unusual spending behavior or transactions with suspicious accounts.

- Cybersecurity: These include detecting anomalies in user activity, e.g., login attempts from different locations or spikes in certain network traffic.

*Personalized Customer Experience*

Real-time data enables businesses to dynamically personalize customer interaction and experience, enhancing levels of customer engagement and satisfaction. For example:

- Retail: Analyzing customer activity on e-commerce platforms for personalized offers and recommendations.

- Media and Entertainment: Streaming platforms use real-time intelligence to make content recommendations or adjust playback quality on the basis of user behavior.

*Operational Optimization*

Industries use the analysis in real-time data streams to gain improvements on operations and resource utilization. For example:

- Supply Chain: Real-time tracking of shipments, levels of inventory, and warehouse operations subsequently allow for the smooth flow of logistics by organizations.

- Energy Management: Utility companies involved in monitoring energy distribution have to constantly keep tabs on power grids and renewable energy sources in real time for this distribution to occur as smoothly as possible.

*Healthcare Applications*

Real-time intelligence plays a pivotal role in improving sentiment analysis, operational performance, and patient outcomes in the healthcare sector. For example:

- Patient Monitoring: Providing continuous monitoring of patient vitals such as heart rate or oxygen saturation helps notify health care providers just before things slightly go off-course.

- Hospital Operations: Monitoring bed occupancy rates and resource use enhances hospitals by optimizing their capacity and improving the care they could deliver.

We have covered a variety of use cases, most of them being industry agnostic. But let's broaden the scope a bit and take a look at how industries can benefit from RTI.

## Real-time Intelligence Industry Applications

Real-time intelligence drastically transforms industries by allowing organizations to analyze and take action on live data streams. By leveraging real-time analytics in operations, industries can achieve more efficient operations, process optimization, and enhancements to customer experience. Real-time intelligence is becoming a staple of modern industry--this section looks at how diverse industries employ real-time intelligence to foster innovation, become more cost-effective, and gain a competitive advantage:

*Manufacturing*

> Real-time intelligence brings a revolution to the manufacturing process by giving near real-time insight into operations while improving quality control and minimizing downtime. For example: A manufacturing unit employs Microsoft Fabric's event streams in the monitoring of their production lines, thus enabling identification of bottlenecks and maintenance of seamless operations in their factories.

*Finance*

> Financial institutions monitor millions of transactions every second to detect fraudulent transactions, the response is near instantaneous, thereby protecting the customers and gaining their trust.

*Retailers*

> These days, retailers and e-Commerce have become dependent on real-time knowledge for efficient customer behavior analytics, inventory management, and pricing strategy optimization. For example, on an e-commerce site, shopping cart activity is tracked live, enabling the

company to offer discounts targeted at that shopper to discourage shopping cart abandonment.

*Transportation and Logistics*

Real-time shipment-trackers and fleet operations help logistics companies optimize their delivery routes and reduce delays. For example, a logistics firm applies live GPS data to reroute trucks in response to traffic conditions or weather-related interruptions.

*Energy and Utilities*

Monitoring energy consumption, achieving optimum power delivery, and effective energy waste reduction are utility firm's primary uses of real-time analytics. In renewable energy, a company might track solar panel output presently, based on demand directing the energy flow.

*Healthcare*

Real-time analytics is used in the healthcare sector to drive better patient outcomes and operational efficiencies. Using a real-time dashboard, hospitals keep track of the admission of patients, the idle resources, and wait times for emergency response.

*Public Sector*

Governments and municipalities harness the insight provided by real-time analytics to watch over city infrastructure, traffic, and public safety. A smart city project takes real-time data from IoT sensors to manage traffic lights and reduce congestion during peak hours.

In summary, real-time intelligence is an enabler for industries ranging from retail and manufacturing to healthcare and finance. Organizations will leverage real-time intelligence capabilities to make decisions, optimize their operations, and deliver better customer service by maximizing the use of data streams. As industries integrate more deeply with digital technologies, the applications of intelligent real-time systems shall continue to proliferate, shaping the future of business and driving innovation across sectors.

## Conclusion

Realizing the difference between BI and RTI is crucial for businesses wishing to access the full potential of their data. Each method serves a different purpose, and when and how to use one can make a big difference in terms of decision-making.

BI enables businesses to identify long-term trends, examine past performances, and derive strategies to meet corporate goals, such as understanding customer buying patterns, analyzing marketing campaign success, and assessing operational effectiveness.

RTI provides real-time insights and continuous analysis of live data streams, allowing businesses to respond quickly to conditions, manage inventory, address security breaches, and make rapid tactical adjustments, which is crucial in time-sensitive industries like e-commerce, finance, and healthcare.

By enabling BI and RTI, you gain a balanced and comprehensive data strategy that combines both approaches' strengths. BI secures decision-making on the basis of a thorough understanding of historical patterns and strategic objectives, while RTI injects nimbleness in terms of reacting to opportunities for benefit or challenges as they pop up. Together they can enable you to navigate a fast-changing market with confidence. The benefits of integration go beyond just those for competitive advantage and responsiveness to customers' needs, operational demands, and market trends. In a world that is quickly becoming data-driven, the power of synergy between BI and RTI is an extraordinarily potent tool towards relevance and innovation.

With this in mind, it's time for action. In the next chapter, we'll take a look at how real-time intelligence works in Fabric so you can start benefiting from data-driven insights.

# Chapter 3. Real-Time Intelligence in Microsoft Fabric

### A NOTE FOR EARLY RELEASE READERS

With Early Release ebooks, you get books in their earliest form—the author’s raw and unedited content as they write—so you can take advantage of these technologies long before the official release of these titles.

This will be the 3rd chapter of the final book. Please note that the GitHub repo will be made active later on.

If you’d like to be actively involved in reviewing and commenting on this draft, please reach out to the editor at *rfernando@oreilly.com*.

As you saw in Chapter 1, Microsoft Fabric integrates data integration, engineering, warehousing, real-time analytics, and business intelligence into a single lake-centric SaaS offering.

Real-Time Intelligence within Microsoft Fabric is a set of capabilities that integrates ingestion, analysis, and visualization of streaming data using a combination of no-code/low-code and full-code alternatives. These capabilities serve various business scenarios by providing unified (near) real-time analytics, direct data ingestion, and scalability of processing. Integrated support for Kusto Query Language (KQL), automated indexing, and partitioning greatly enhance data handling in Fabric. It has automation triggered by the arcs that enable immediate responses to critical events while ensuring built-in security for high-level data protection. The allows no-code/low-code experiences to enable those with differing technical expertise to adopt real-time analytics, providing agility and well-informed decision-making across teams regardless of competencies.

### REAL-TIME VERSUS NEAR REAL-TIME

Before we continue with the book, let us first clarify what real-time is and why we in this book prefer to talk about *near* real-time.

When we deal with events in streams, most often we call this real-time - but as we send events away from the event producer, we introduce a latency. So whenever we send data from say, a sensor on an engine up to a cloud data processing system such as Microsoft Fabric, we will have to wait for up to several seconds after the event occurs until we can take action on it. And the more processing we do of the event, the more latency we introduce.

So for proper real-time analytics the data is being handled on the factory floor - or in the field so to speak.

What we deal with in Microsoft Fabric Real-Time Intelligence is always *near* real-time. By using a cloud service, we accept that we have a latency - but how large that latency will be is something that we can find out with testing.

## 3.1 The Components of Realtime Intelligence

Microsoft Fabric Real-Time Intelligence is a total ecosystem engineered to get actionable insights from streaming data. This gives shape to several key components running in seamless unison. You can see an overview and how the different components relate in Figure 3-1.

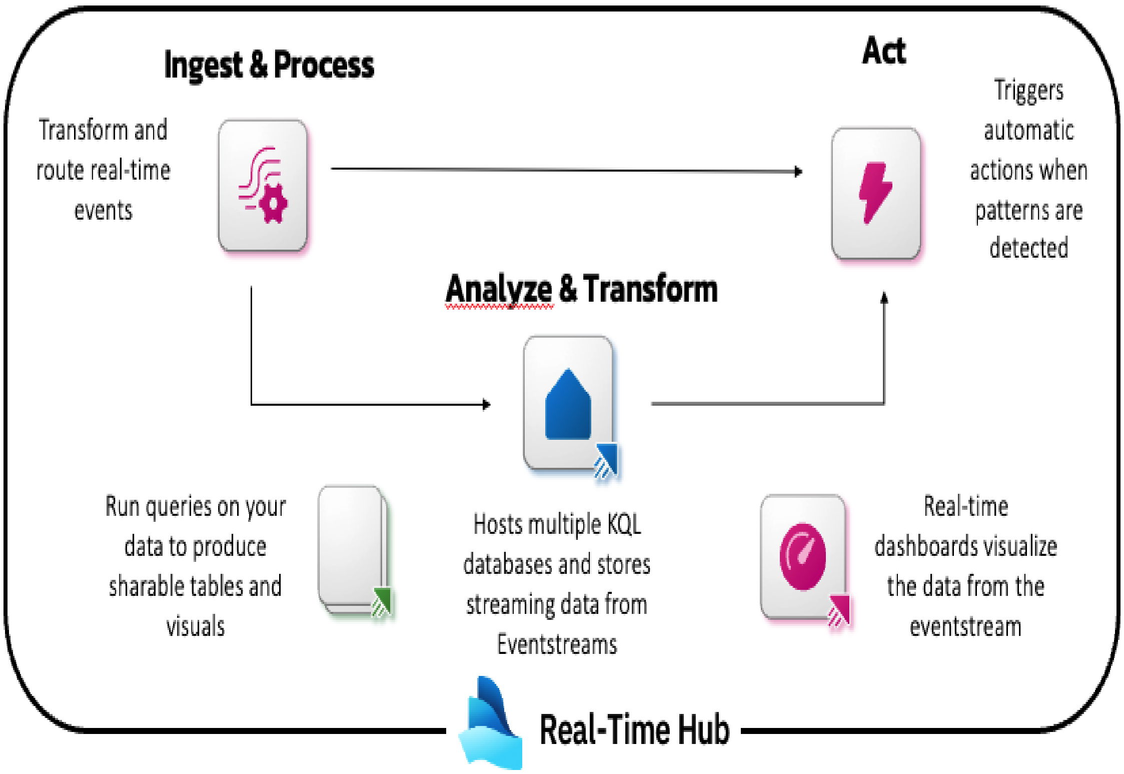



*Figure 3-1. Overview of how the different components of Real-Time Intelligence relate to each other*

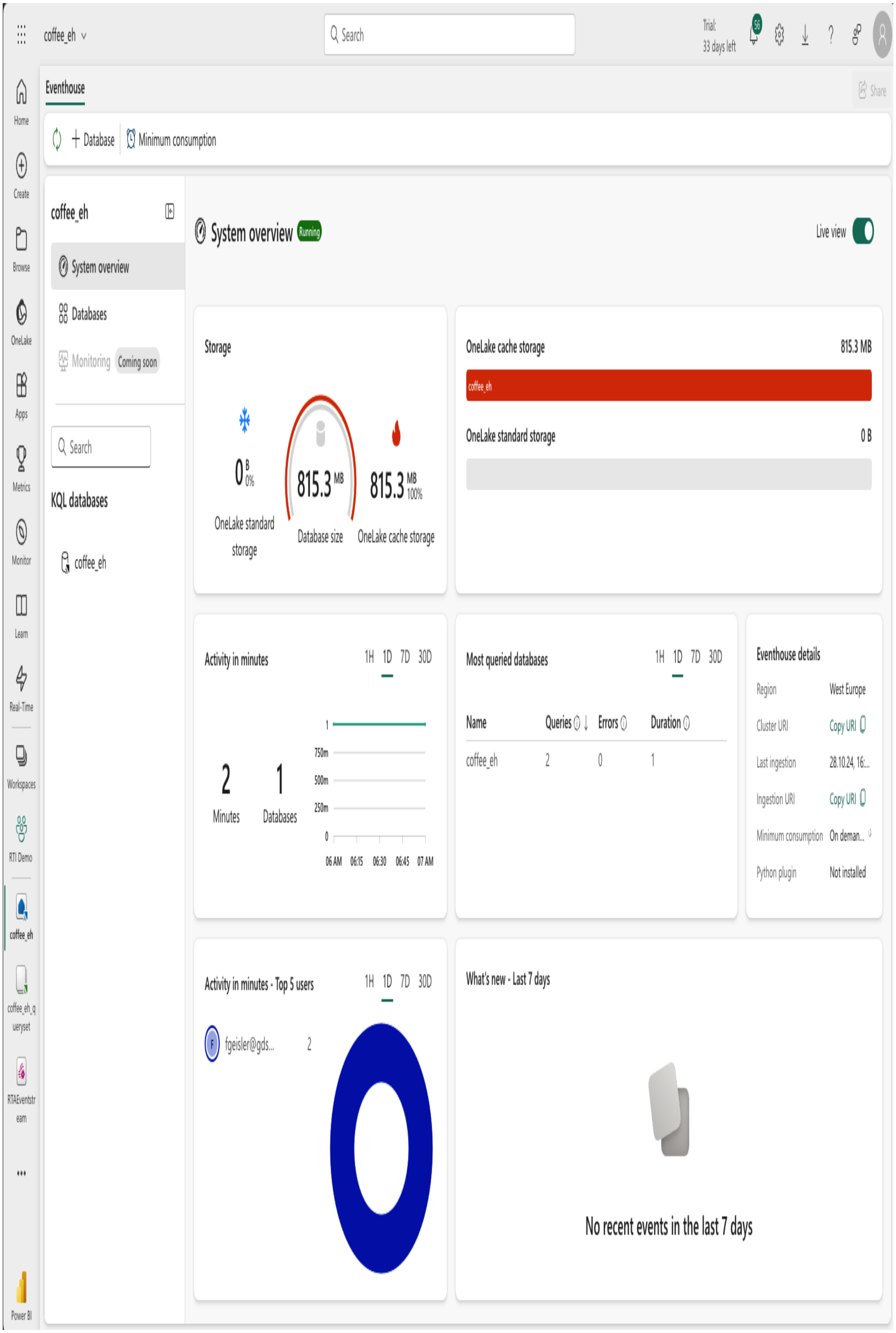
In this chapter you will learn how the Real-time Intelligence service in Fabric can help you ingest and process, analyze and transform, and act upon data to drive actionable insights and decisions.

Let's go through each of the components in detail.

## 3.1.1 Eventhouses

Eventhouses are an intrinsic part of the Real-Time Intelligence of Microsoft Fabric that cache and store time-series data.

You can see an Eventhouse in the main view of Fabric, shown in Figure 3-2.



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Search
Trial:
33 days left
Eventhouse
Share
Database
Minimum consumption
Home
Create
Browse
OneLake
Apps
Metrics
Monitor
Learn
Real-Time
Workspaces
RTI Demo
coffee_eh
coffee_eh_queryset
RTAEventstream
Power BI
coffee_eh
System overview
Databases
Monitoring
Coming soon
Search
KQL databases
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System overview
Running
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815.3 MB 100%
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OneLake cache storage
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OneLake standard storage
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Minutes
1
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750m
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07 AM
Most queried databases
1H 1D 7D 30D
Name
Queries
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1
Eventhouse details
Region
West Europe
Cluster URI
Copy URI
Last ingestion
28.10.24, 16:...
Ingestion URI
Copy URI
Minimum consumption
On deman...
Python plugin
Not installed
Activity in minutes - Top 5 users
1H 1D 7D 30D
fgeisler@gds...
2
What's new - Last 7 days
No recent events in the last 7 days

*Figure 3-2. The System overview of an Eventhouse*

Within an Eventhouse, you will have a collection of KQL databases and KQL querysets.

> **NOTE**
>
> As KQL databases provide the actual capability for ingesting, storing and serving the time-series data, we won’t refer to Eventhouse in that capacity..

The hierarchy between Eventhouses, KQL-Databases and KQL Querysets is as follows: One Eventhouse can have one or many KQL-Databases while each KQL-Database can have one or many KQL Querysets. This can be seen in Figure 3-3.

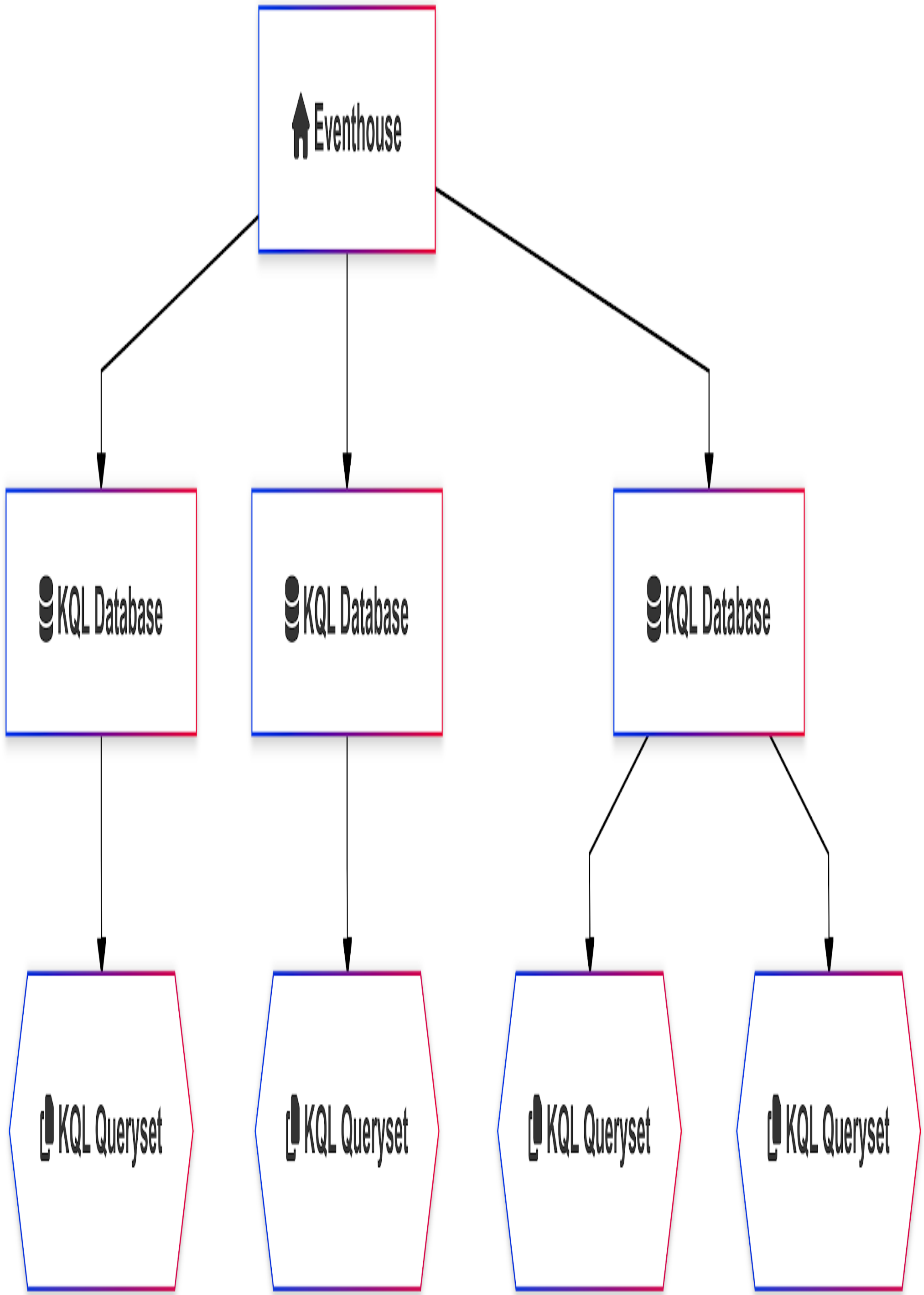

Eventhouse
KQL Database
KQL Database
KQL Database
KQL Queryset
KQL Queryset
KQL Queryset
KQL Queryset

*Figure 3-3. Hierarchy between Eventhouse, KQL Databases and KQL Querysets*

Let's work our way down.

## 3.1.2 KQL Databases

KQL databases can handle huge quantities of time-series data ingested from different sources such as IoT devices, telemetry systems, and application logs. As more organizations depend on real-time insights to shape their operations, the Eventhouse and it's collection of databases comes with the infrastructure to ingest, process, and retain data at scale.

The key competencies of KQL databases are:

*High Volume Data Handling*

> KQL databases are well suited to handle time-series data complexities at high volume. This includes data of all different natures, such as IoT sensors monitoring conditions on industrial equipment or telemetry logs tracking performance for the application.

*Integration with Varied Sources of Data*

> KQL databases are able to integrate with a very broad set of data sources. The KQL database accepts connections to platforms like Event Hub, Kafka, Amazon Kinesis, Google Pub/Sub as well as other SDKs for backend integration, hence enabling organizations to create a one-stop data pipeline from various sources. That versatility means KQL databases could plug into just about any data architecture while operating in a unified repository for real-time analytics.

*Automatic Indexing and Partitioning*

> KQL databases automatically do all indexing and partitioning, with a view towards seamless querying performance. Because of how it's organized on ingestion, KQL databases generate faster responses to queries-even with petabytes of data. These become really important to applications that are sensitive to time, with the variance of speed

determining the distinction between proactive and reactive decision-making.

*Unified Database Management*

Eventhouse enables unified KQL database management, offering a single workspace for database performance tracking, access control configuration, and data retention for all KQL databases.

*Built-in Monitoring*

Eventhouse's monitoring utility offers real-time insights on KQL database health, query performance, and trends for data usage. These allow administrators to pinpoint bottlenecks, optimize resource allocation, and maintain data availability. In addition, Eventhouse facilitates proactive KQL monitoring so that a consistently high level of performance and reliability can be maintained along the entire real-time data pipeline.

As you can see, by providing a centralized evented workspace for real-time data management, an Eventhouse and its contained KQL databases allow you to take full advantage of their time-based modeling data and make smart data-driven decisions. To interact with that real-time data, you need an interface: KQL Queryset.

### 3.1.3 KQL Queryset

Where KQL Databases are the core of the Real-Time Intelligence experience on Microsoft Fabric, KQL Querysets enable users to work with the data within the Fabric UI. It combines KQL power with SQL friendliness and dynamic results customization and visualization. As such KQL Queryset is a great tool for developers and data analysts wanting to do timeseries and real-time analytics efficiently.

The benefits of using KQL Querysets are:

*A powerful query language*

The main selling point of KQL is its ability to query real-time data sources like telemetry logs, events and other timeseries data. You are able to gain insights on huge amounts of data within seconds. The rich syntax support of KQL means queries may range from simple lookups to pattern detections and complex aggregations.

*Arrange Queries in the Way You Want*

You can sort, filter, and format the data the way tiy prefer. Results can be visualized directly within the tool, with time charts, bar graphs, and other visualizations without ever moving out of the query interface.

*SQL-Compatible*

Even though KQL is the native language of the KQL databases, you can also use SQL to bridge the gap between the traditional users of relational databases and those entering into real-time analytics. Dual support allows users of different levels of technical ability to maximize existing SQL competence, while easing into KQL's more powerful analytical query functionality.

*Automatically saved queries*

The ability to have all queries automatically saved, and then being able to name and share them is common to all query editors in Microsoft Fabric. This functionality eases tasks by enabling users to reuse predefined queries instead of having to write them every time. This feature is beneficial for routine jobs like monitoring or generating standardized reports.

*Functions and Operators*

KQL Querysets have several options of functions and operators that makes query construction easier. For instance, these include arithmetic operators for calculations, comparison operators for filtering, logical operators for complex conditions, and string manipulation tools for text processing.

*Advanced Queries*

Features such as aggregations, joins on tables, and subqueries lead to more advanced exploratory analysis or manipulation of data. With such features, you can cross-combine data from several different tables, compute statistical summaries or dynamically filter the results, all making for deeper explorative analysis into user datasets.

*Export Options*

You can export data into different formats such as CSV and Excl or integrate the results into downstream systems and workflows.

*Multiple Tabs for Query Contexts*

Each of the tabs works independently, allowing the user to store a separate query context for various datasets. This is similar to the experience in Management Studio and Azure Data Studio, and makes it easy to keep track of various queries against multiple databases.

*Query Sharing Features*

You can share queries with relevant parties via Teams, E-mail, or a direct link.

*Quick and Efficient*

KQL is engineered for analytics over petabyte scale datasets.

*User-Friendly*

KQL Querysets support syntax highlighting and Intellisense.

To get to the KQL Queryset user interface in Fabric, go to an Eventhouse and expand the KQL-Databases that are assigned to this Eventhouse. Within one of the KQL-Databases expand the Node KQL Queryset. All Querysets that are defined for this KQL Database will be shown. If you click on one of these you will see the interface shown in Figure 3-3.

The key components of the KQL Queryset user intefaces are:

*Query Editor*

The query editor enables creation, modification, and execution of queries in KQL from the user interface of Microsoft Fabric. It incorporates features like syntax highlighting, IntelliSense for auto-completion, and inline error validation. The editor makes it easy to write queries. Its built-in run functionality allows queries to be executed instantly with results shown in the results pane below the query editor. Additionally, saving a query is automatic, and you can share the queries, which makes it easy to collaborate with others.

*Tabs for multiple query contexts*

The Tabs for Multiple Query Contexts allows users to work on different datasets from a single interface. As each tab functions independently, you can have the capability to access different KQL databases at the same time and execute queries limited to their data context. This experience feels familiar to users used to working with Management Studio or Data Studio and SQL Server flexible, familiar and easy. Moving between tabs is easy and you can rename the tabs as well as provideing different database connections per tab.

*Database Explorer*

The Database Explorer makes it easy for you to quickly find databases, schemas and tables you need in your queries. The Database Explorer allows you to preview data, retrieve table schemas, or generate sample queries - such as top 100 - that can be executed directly from the interface.

*Query Actions Toolbar*

The Query Actions Pane in KQL Queryset UI gives you quick access to functionality such as start and stop queries, clear out query results, exporting query results or pin visualizations.

*Results Pane*

The results pane shows query r the results of queries. Results are displayed in tabular format, which makes it easy to read. The pane gives you the option to sort, filter, pivot and export results to CSV or Excel for deeper analysis or share them with collaborators.

*Sharing Options*

The Sharing Options allows you to share your work through various means, such as allowing users in your organization to see or edit the query, share a link to the query via a URL, send it as an email, or share on Teams.

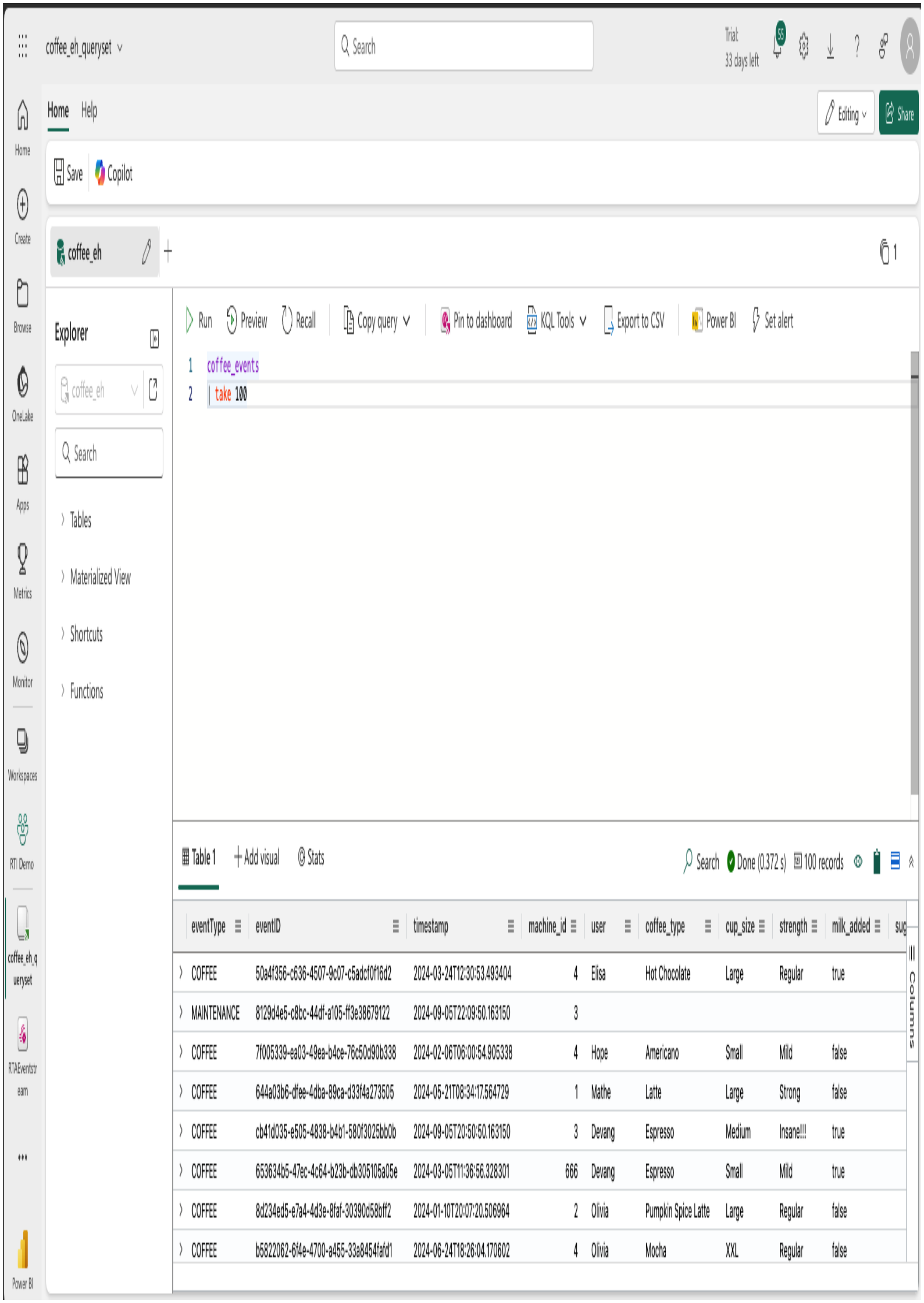



*Figure 3-4. A KQL-Queryset with a KQL Query and result Data.*

As we have seen in this section how time based data is stored lets move to the next section where you will learn which component is able to ingest a stream of data into the KQL Databases.

### 3.1.4 Eventstream

Microsoft Fabric's Eventstream is a low-code experience that permits users to ingest, process and route real-time event data. The interface visually reduces the complexities surrounding event data handling. With its multi-format support and integrated tools, Eventstream makes real-time streaming analytics accessible for more users.

Figure 3-5 shows the design GUI for creating Eventstreams.

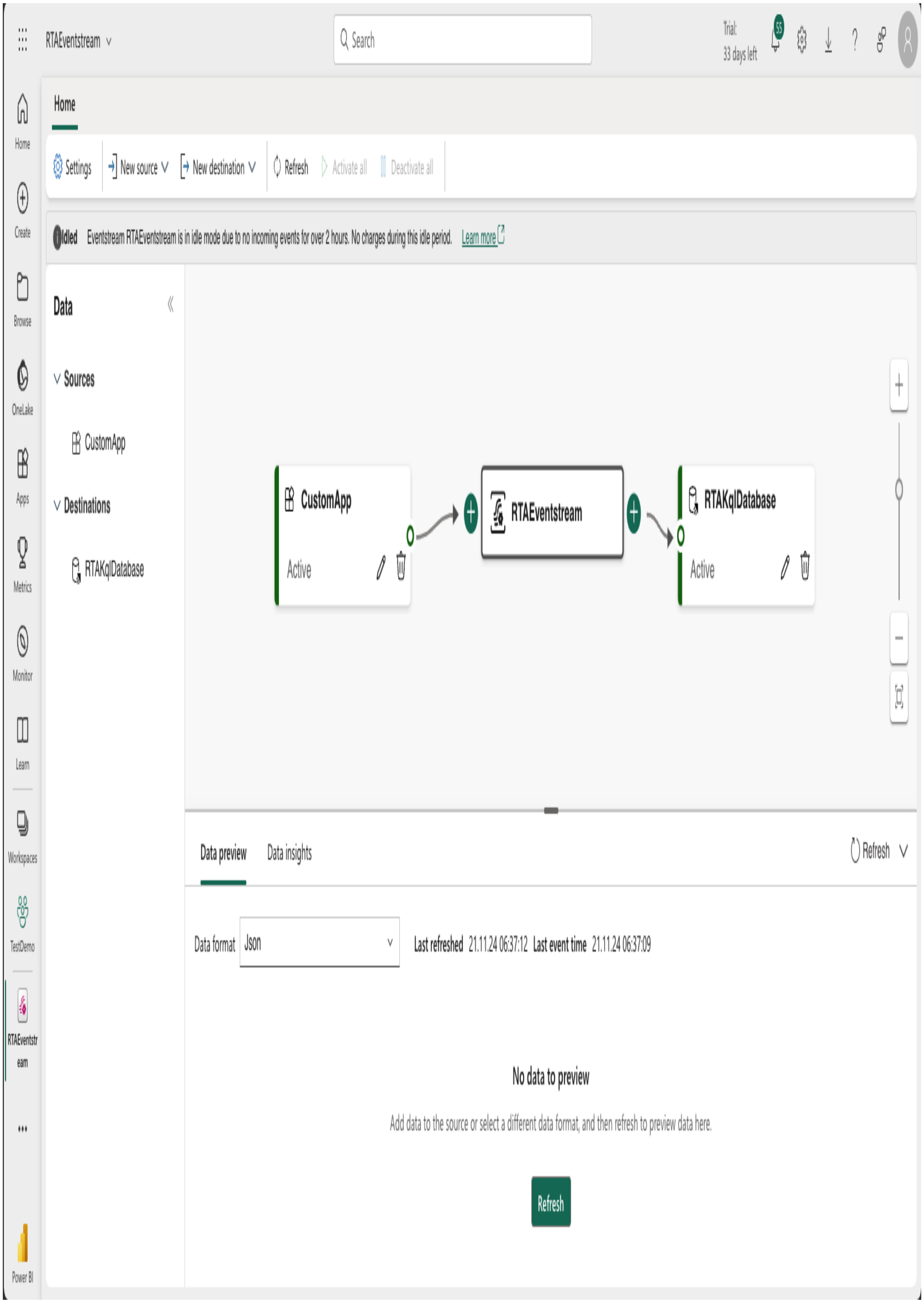



*Figure 3-5. The GUI for creating Eventstreams*

The key components of the Eventstream canvas, or designer, are:

*Main Canvas*

The canvas - or designer - of Eventstream is a visual no-code workspace where we can drag various functions and components and drop them onto the canvas. This canvas supports different data sources, including Azure Event Hubs, IoT Hubs, and custom applications, along with various destination options such as Lakehouses, KQL databases and APIs, among others. The eventstream designer gives you a preview of the data you ingest which allows you to understand the schema of the events coming in. You can then add transformations and send data to multiple downstream targets from within the canvas.

*Ribbon Menu*

The Ribbon Menus give you access to the context-sensitive tools and operations of workflow designs. This menu is logically laid out, so that frequently used items are most reachable, thus reducing the need for complicated navigation. Overall, a ribbon menu centralizes the most important controls, making the developer's tasks easier.

*Configuration and Modification Pane*

The Configuration and Modification Pane is where you can customize and tune event stream components. It provides a control area where the detailed properties of the data source, event processors, and destinations can be configured. Be it defining several transformation settings, windowing functions such as hopping or tumbling, or output format management, the interface gives you easy access.

*Information and Logs Pane*

The Information and Log Pane is where you monitor and debug the Evenstreams. Positioned at the bottom half of the interface, it provides insights about the performance and health of the event stream, such as runtime logs, previews of data, error messages, and selected key metrics that help the user to quickly diagnose and fix issues. The pane gives insights on miscellaneous data to such an extent as record counts and different statuses of processing, enabling real-time validation and optimization of workflows.

*Input Sources*

Input Sources are - naturally - at the heart of our event streams, and support the integration of multiple data feeds. Users can choose from several supported sources such as Azure Event Hubs, Azure IoT Hubs or Kafka and custom applications. Each one is quick to configure. AMQP and HTTPS are standard protocols supported, enhancing flexibility and connectivity. Eventstream also enables users to preview incoming data with a buffer for up to thirty days, ensuring sources are configured correctly before use.

*Transformations*

In an Eventstream you can use Transformations on the streaming data as it streams in. Available Transformations are: Filter, Manage Fields, Aggregate, Join, Group by, Union and Expand. These Transformations can be used to bring the data into shape before they are stored in the Data Destination. The offered transformations are somewhat basic. If you want to do complex transformations you can do them in the database instead.

*Data Destinations*

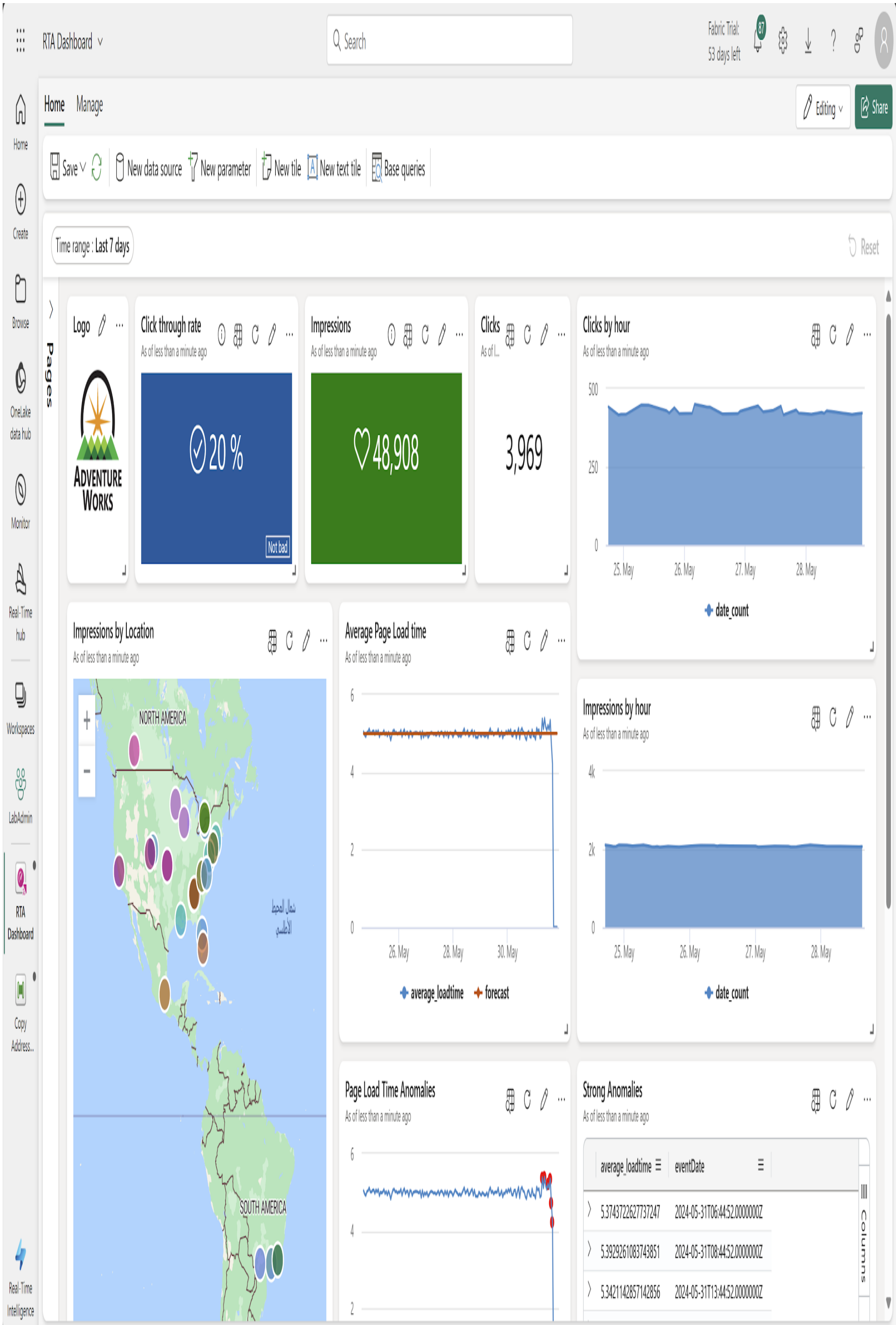
These are the destinations the data will be routed to. A data destination could be either a data storage that will store the data or an activator that will act on the data.

Eventstream consists of source selection, configuring event processors, and routing over to appropriate destinations. Everything is code-free so the user doesn't have to know a specific code language to use the service.

### 3.1.5 Real-Time Dashboard

When working with streaming data, and near real-time insights it is important to use the tools that introduce the least latency in the end-to-end stream, from source to target. And while Power BI is a great visualization and analytics tool, it wasn’t designed for near-real time analytics. Data might seem continuously updated in Power BI Desktop, but as soon as you deploy you will notice the changes. And how about filtering on the latest data?

Fabric RTI addresses these issues with Real-Time Dashboards. Real-Time dashboards deliver low latency visualization of your KQL queries, and while Power BI looks prettier we do get the visualizations we need to get the message across, and some tools to prettify the layout. Easy to set up and configure, and refreshing continuously - the Real-Time Dashboards fill a gap in the Microsoft reporting-stack that some of us have been missing for a while. An example for a Real-Time dashboard can be seen in Figure 3-6.



RTA Dashboard
Search
Fabric Trial:
53 days left
Home
Manage
Editing
Share
Save
New data source
New parameter
New tile
New text tile
Base queries
Time range : Last 7 days
Reset
Create
Browse
OneLake data hub
Monitor
Real-Time hub
Workspaces
LabAdmin
RTA Dashboard
Copy Address...
Real-Time Intelligence
Pages
Logo
ADVENTURE WORKS
Click through rate
As of less than a minute ago
20 %
Not bad
Impressions
As of less than a minute ago
48,908
Clicks
As of l...
3,969
Clicks by hour
As of less than a minute ago
500
250
0
25. May
26. May
27. May
28. May
date_count
Impressions by Location
As of less than a minute ago
NORTH AMERICA
SOUTH AMERICA
شمال المحيط الأطلسي
Average Page Load time
As of less than a minute ago
6
4
2
0
26. May
28. May
30. May
average_loadtime
forecast
Impressions by hour
As of less than a minute ago
4k
2k
0
25. May
26. May
27. May
28. May
date_count
Page Load Time Anomalies
As of less than a minute ago
6
4
2
Strong Anomalies
As of less than a minute ago
average_loadtime
eventDate
5.374372262737247
2024-05-31T06:44:52.0000000Z
5.392926108374385
2024-05-31T08:44:52.0000000Z
5.342114285714285
2024-05-31T13:44:52.0000000Z
Columns

*Figure 3-6. A good example for a Real-Time Dashboard*

Key Features of Real-Time Dashboards are:

*Real-Time Optimized Visualizations for Streaming and Event Data*

Real-time dashboards are typically optimized for low-latency, high-velocity streaming data and allow dynamic visualizations through the use of charts, graphs, and heatmaps. They are most often appropriate for monitoring hardware such as IoT devices, managing financial transactions, and analyzing system logs.

*Integration into the Power BI*

These are not standalone dashboards but integrate with Power BI seamlessly. Thus, it allows organizations to leverage their existing capabilities of analytics and reporting by providing every user with a consolidated interface for operating real-time and historical data analysis.

*Low-Latency Refresh*

Real-time dashboards, backed by the compute engines and event-processing features of Microsoft Fabric allow low-latency updating of dashboards.

*Powered by KQL*

Kusto Query Language is the querying technology of the real-time dashboard. KQL is easy to use and quick. This makes it very convenient for COBIT testing and simulation purposes.

*Parameterized Queries*

The dashboards will provide high degrees of customization by using parameterized queries. Users can filter or rework the visualization directly, without changing the underlying queries. For example, an operations manager can modify a time range or configure a selection of the area in the visualization.

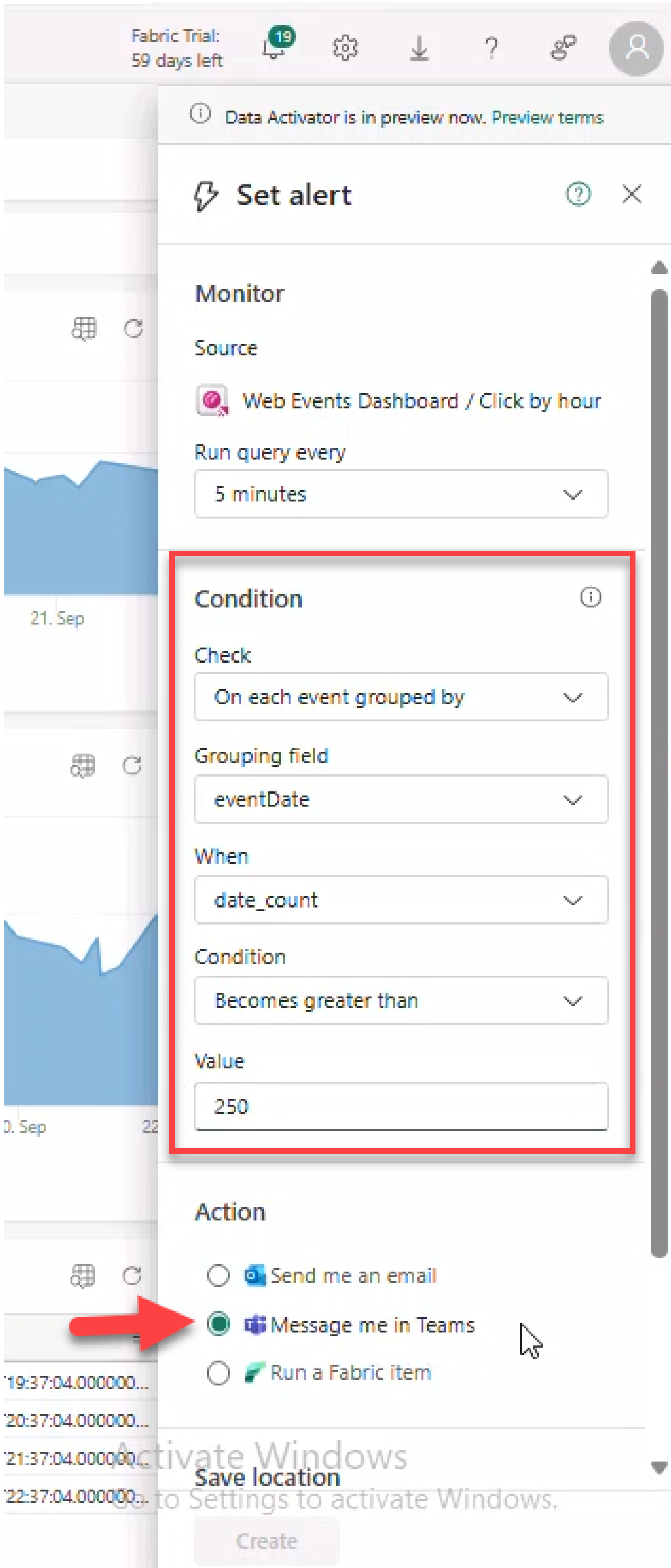
*Cross-Interaction and Filter*

> The real-time dashboards allow visual component cross-filtering and interaction. Selecting one or more data points in a visualization dynamically updates related visualizations, thus allowing for an interactive exploration of the dashboard.

Real-time dashboards supplement Power BI, filling the gap between Power BI's strength of offering self-service analysis and visualization of historical and aggregated information. Whereas real-time dashboards expand on this by offering low latency time series analytics and visualization. Real-time dashboards are a great addition to the Real-Time Intelligence capabilities of Fabric.

### 3.1.6 Activator

Activator is a low-code, no-code way in Fabric to take action on data in real-time. When Fabric was first introduced Activator, which was called Reflex that time, was tagged as one of the core experiences on the platform on the same level as Real-Time Analytics and Data Engineering to mention two others. When Fabric went Generally Available (GA), Real-Time Analytics and Activator were joined together to become Real-Time Intelligence. This makes sense from several perspectives, especially as a core functionality when handling streaming data is to be able to take action on the events as they happen. In Figure 3-6 you can see the side pane where you can configure an alert on change of data. Technically an activator will be created out of your alert configuration.



Fabric Trial:
59 days left
19
Data Activator is in preview now. Preview terms
Set alert
Monitor
Source
Web Events Dashboard / Click by hour
Run query every
5 minutes
Condition
Check
On each event grouped by
Grouping field
eventDate
When
date_count
Condition
Becomes greater than
Value
250
Action
Send me an email
Message me in Teams
Run a Fabric item
Save location
Create
21. Sep
Activate Windows
Go to Settings to activate Windows.

*Figure 3-7. The Side-Pane where you can configure your Activator.*

Key Features of Activator are:

*Actionable Insights with Real-Time Notifications*

Activator lets users determine triggers and conditions to either notify or automatically respond through predefined actions upon occurrence of specific events. Businesses may trigger alerts for events such as “low inventory” or “excessive network traffic”.

*No-Code Setup*

Activator simplifies the entire setup process with a low-code/no-code interface that allows users with no programming experience to create sophisticated workflows.

*Focus on Business Processes*

Activator is designed with a focus on business workflows, allowing users to create actions that trigger when an event happens or a condition is met, creating Teams notifications about delayed shipments or escalating critical system outages to mention a few examples.

*Integration with Real-Time Dashboards and Power BI*

Activator works hand-in-hand with real-time dashboards and Power BI allowing users to define triggers to send information messages or take actions.

*Centralized Control*

A centralized control interface where users can manage rules, triggers, and actions. While users have context-aware trigger creation in various parts of the Real-Time Intelligence experience - such as in Real-Time Dashboards or KQL Querysets, the centralized Activator interface allows for further customization as well as management, monitoring and updates.

*Sample Data and Tutorials*

> Activator provides sample data along with guided tutorials to enable quick onboarding and application of solutions fitting for the user. Some offer an easier learning curve, encouraging faster adoption.

Activator enables proactive intelligence on organizations' data streams to gain automatic triggering of workflows or alerting the right people. With a no-code interface and centralized control, Activator is a robust solution that is not only flexible, but reduces the need for specially trained developers to create and maintain the solutions.

## 3.3.6 The Real-Time Hub

Microsoft Fabric Real-Time Hub acts as a centralized interface for real-time data stream management, analytics, and decision-making. It is the data catalog for all Real-Time Intelligence related assets, it lists all event streams, tables, Eventhouses, KQL databases and Realt-Time Dashboards. To get to the Real-Time hub just click on the Real-Time icon in the left pane. If you do so, the Real-Time hub will be shown as you can see in Figure 3-8.

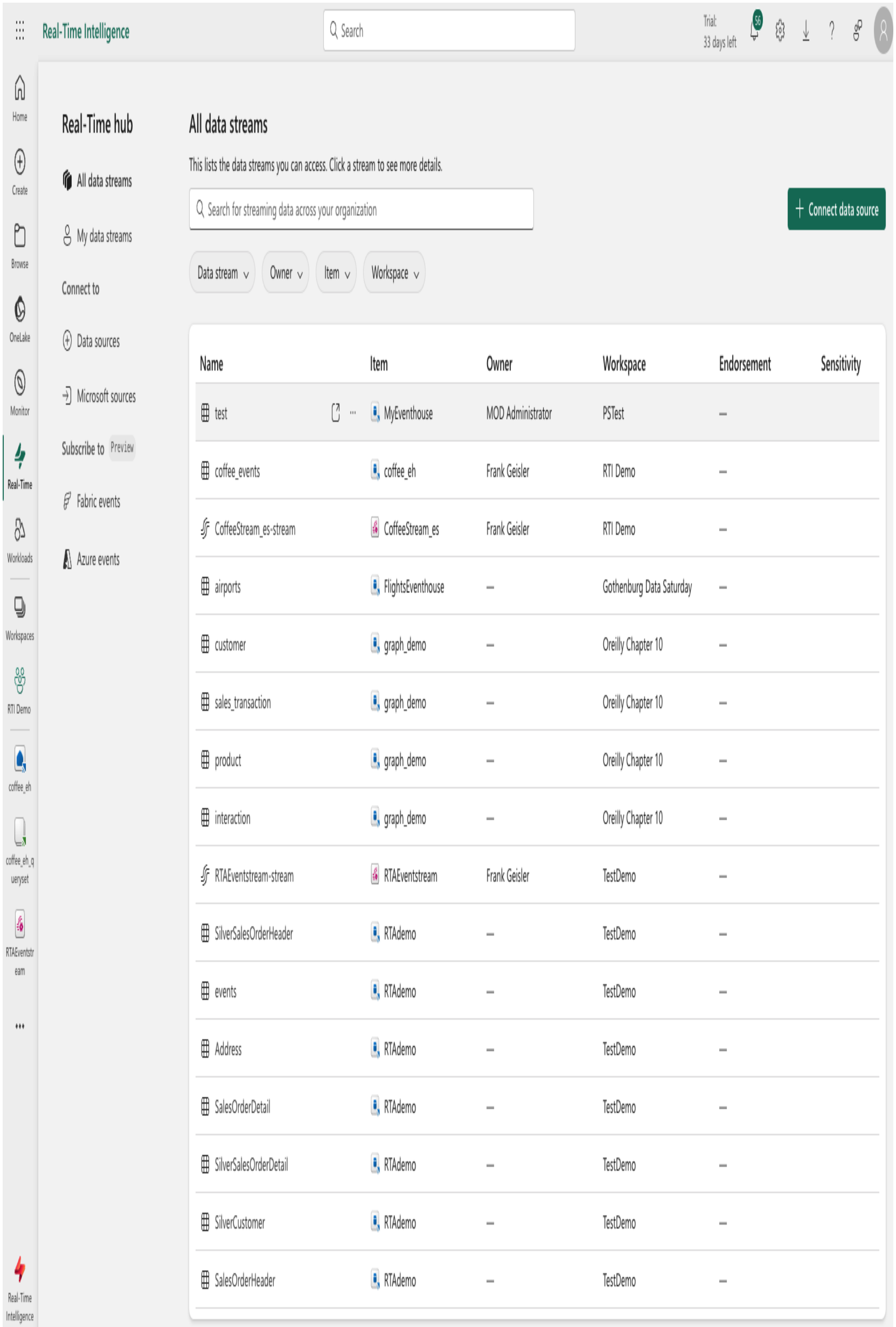

Real-Time Intelligence
Search
Trial:
33 days left
Home
Create
Browse
OneLake
Monitor
Real-Time
Workloads
Workspaces
RTI Demo
coffee_eh
coffee_eh_queryset
RTAEventstream
Real-Time Intelligence
Real-Time hub
All data streams
My data streams
Connect to
Data sources
Microsoft sources
Subscribe to Preview
Fabric events
Azure events
All data streams
This lists the data streams you can access. Click a stream to see more details.
Search for streaming data across your organization
Connect data source
Data stream
Owner
Item
Workspace
Name Item Owner Workspace Endorsement Sensitivity
test MyEventhouse MOD Administrator PSTest —
coffee_events coffee_eh Frank Geisler RTI Demo —
CoffeeStream_es-stream CoffeeStream_es Frank Geisler RTI Demo —
airports FlightsEventhouse — Gothenburg Data Saturday —
customer graph_demo — Oreilly Chapter 10 —
sales_transaction graph_demo — Oreilly Chapter 10 —
product graph_demo — Oreilly Chapter 10 —
interaction graph_demo — Oreilly Chapter 10 —
RTAEventstream-stream RTAEventstream Frank Geisler TestDemo —
SilverSalesOrderHeader RTAdemo — TestDemo —
events RTAdemo — TestDemo —
Address RTAdemo — TestDemo —
SalesOrderDetail RTAdemo — TestDemo —
SilverSalesOrderDetail RTAdemo — TestDemo —
SilverCustomer RTAdemo — TestDemo —
SalesOrderHeader RTAdemo — TestDemo —

*Figure 3-8. The Real-Time hub shows all Real-Time assets you can access*

The core capabilities of the Real-Time Hub are:

*One-Stop shop for users looking for streaming data*

In the Real-Time Hub, users can easily find all data streams that they are allowed to see. They can see which streams are verified and which are promoted. And they can share data streams to the hub to allow other users to build new insights without having to “reinvent the wheel”.

*Centralized Data Stream Management*

The Real-Time Hub offers a single interface for managing the integration of various data sources, be they IoT systems, Azure Event Hubs, or custom applications. This centralization helps simplify the complexity of event data pipelines.

*Eventstream Integration*

It can integrate with the Eventstream module inside Microsoft Fabric to allow users to create workflows to capture, transform, and route event data. Native connectors and a no-code canvas simplify the creation of event-driven processes for making their lives easier.

*Alerts and Actions with Activator*

Activator is a ‘critical element’ in the Real-Time Hub, providing automation to trigger notifications and predefined actions. It is specifically useful for various scenarios such as anomaly detection, proactive maintenance, and process automation.

The Microsoft Fabric Real-Time Hub gives users an overview over all real-time intelligence assets in the organization. It offers a centralized spot for ingesting, processing, and acting on live data so that organizations can respond to changing conditions robustly and proactively.

## 3.4 Conclusion

Microsoft Fabric serves as an integrated platform for real-time intelligence for businesses. Fabric, through Eventstreams for data ingestion, KQL databases for analytics, Reflex for automation, and real-time dashboards for visualization simplifies the real-time intelligence platform for organizations.

In this chapter we showed you the components that make up the Real-Time Intelligence experience in Fabric.

Eventhouse is the central storage for real-time data management. Eventstream ingests, transforms, and routes high-velocity event data from multiple sources. KQL Querysets provide an interactive analysis and visualization environment for data-using the mighty Kusto Query Language - KQL, supporting both ad-hoc and extensive analyses. Real-Time Dashboards dynamically visualize live data, empowering users to make instantaneous data-driven decisions. Finally, Reflex (Data Activator) automates actions for whenever a specific condition is met. These features comprise a strong platform for real-time analytics that enables organizations to stay agile and responsive in today's fast-paced, data-driven world.

Let's summarize and have a look at the different areas of an analytical solution and how these relate to the different parts of Real-Time Intelligence.

*Ingest & Process*

> The start involves the changing and routing of live events by means of Eventstreams, assuring that data from the varied sources is captured, cleansed, and then routed to the proper destinations for further cleansing, transformation, and analysis.

*Analyze & Transform*

> At this stage, the Real-Time Hub contains multiple KQL databases for streaming data. With KQL, users can run queries, analyze patterns, and produce sharable tables and visuals. Those would become the basis of real-time decision-making.

*Act*

> Upon detection of data patterns or anomalies, Activator triggers the automated actions. Those could be anything from sending critical alerts, to starting workflows, or executing pre-established responses, thus guaranteeing encouragement to business events.

*Visualize*

> Real-time dashboards give a dynamic interface to visualize the streams of events and the insights drawn from the analysis. The dashboards keep stakeholders informed and engaged with modern updates and their work.

The next chapter is dedicated to building a complete real-time intelligence solution from scratch to finish. This voyage will carry you through the processes involved in setting up data ingestion and transformation through Eventstreams and analyzing real-time data using KQL databases.

## About the Authors

**Johan Ludvig Brattås** is a director at Deloitte, and a dedicated community guy. He has worked with MS SQL server since late 1999, mostly with BI in one form or another. Since 2015, most of his work has been in the cloud working on data platform services such as Snowflake, Databricks and Synapse.

Combining his passion for MS SQL Server with his passion for sharing knowledge, he started speaking at various events in the SQL Community. This is also a way to give back to the community for all the things he has learned over the years. When not working, Johan Ludvig spends his time with his kids, playing with new technology or teaching coeliacs how to bake gluten-free food.

**Frank Geisler** is owner and CEO of GDS Business Intelligence GmbH. He is Data Platform MVP, MCT, Azure Solutions Architect Expert, Azure Security Engineer Associate, Azure Data Engineer Associate and DevOps Engineer Expert In his Job he is building Business Intelligence Systems based on Microsoft Technology, mainly on SQL Server and Power BI and he is also building Azure Infrastructures and Architectures with PowerShell. Besides this he is author of several books including Power BI for Dummies, Azure for Dummies, Docker for Dummies and Pro Serverless Data Handling with Microsoft Azure. He is also a frequent speaker at national and international Conferences including PASS Summit, SQL BITS and SQL Server Konferenz. In 2004 he was cofounder of PASS Deutschland e.V. where he is at the board of directors for many years and he runs the Microsoft Data Community Regional Chapter Münsterland.